

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	AIML	Date:	18/08/2028

Code of Conduct

I N.Adi Narayana (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ Adi _____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.

- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Given the following paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability)
- Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. Conversation History: User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

Constraints:

1. Maintain **entity coherence** (reuse "exam", "concentrate", "you").
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. **Source Sentence:** “The bank by the river flooded after the storm, but it was saved by quick action.”

Note: “bank” is ambiguous (financial institution or riverbank).

Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

Tasks: a) Resolve the ambiguous word “bank” and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

Q1. Discourse, Reference Resolution and Text Coherence

Given Paragraph

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

The objective is to identify referring expressions, determine their antecedents using constraints, construct a constraint graph/table, generate final coreference chains, and justify the priority of the constraints. The specified constraints are gender and number agreement, recency, semantic compatibility, and coherence.

a) Referring Expressions and Possible Antecedents

A referring expression is a word or phrase that refers to an entity previously introduced in the discourse.

The important referring expressions in the paragraph are:

1. He
2. She
3. it
4. him
5. they
6. all

Identification

Referring Expression	Possible Antecedents	Final Antecedent
He	John, Mary	John
She	John, Mary	Mary
it	ball, park, John, Mary	ball
him	John, Mary, dog, ball	John
they	John, Mary, dog	John + Mary + dog
all	John, Mary, dog	John + Mary + dog

"He"

Sentence: "He brought a ball."

The possible human antecedents are John and Mary. The gender constraint eliminates Mary because "He" is masculine.

Therefore: He $\rightarrow$ John

"She"

Sentence: "She wanted to play with it."

The possible antecedents are John and Mary. The gender constraint eliminates John because "She" is feminine.

Therefore: She $\rightarrow$ Mary

"it"

Sentence: "She wanted to play with it."

The previously introduced objects include the park and the ball. The semantic compatibility constraint is important here. Mary can play with a ball, but play with the park is semantically inappropriate.

Therefore: it $\rightarrow$ ball

"him"

Sentence: "The dog chased him excitedly."

Possible antecedents include John, Mary and the dog itself. The masculine pronoun "him" cannot refer to Mary. The noun dog is also a poor antecedent because the sentence would result in the dog chasing itself. Semantic compatibility strongly supports him $\rightarrow$ John. A dog can naturally chase a person.

"they"

Sentence: "Finally, they all went home."

The plural pronoun "they" refers to the group consisting of John + Mary + dog.

Therefore: they $\rightarrow$ John, Mary and dog

b) Applying the Constraints

The four specified constraints are:

1. Gender and number agreement
2. Recency
3. Semantic compatibility
4. Coherence

1. Gender and Number Agreement

Pronouns must agree with their antecedents in grammatical features. Examples: He is masculine singular, She is feminine singular, it is singular/neuter, they is plural. Therefore He $\neq$ Mary and She $\neq$ John. This is the strongest initial filtering mechanism.

2. Recency Constraint

The recency principle prefers an antecedent that was mentioned recently. For example, when processing "The dog chased him excitedly," the system examines recently mentioned entities first. However, recency alone cannot solve the problem. The immediately preceding entity is dog, but dog $\rightarrow$ him would be semantically inappropriate because the dog cannot naturally chase itself in this context. Therefore, recency should be combined with semantic compatibility.

3. Semantic Compatibility

Semantic compatibility checks whether the proposed antecedent makes sense with the action. "Mary wanted to play with the ball" is semantically valid, but "Mary wanted to play with the park" is not a natural interpretation. Similarly, "Dog chased John" is semantically reasonable, but "Dog chased itself" is not the intended interpretation.

4. Coherence

Coherence requires the same entity to maintain a consistent identity throughout the discourse. For example, He $\rightarrow$ John and him $\rightarrow$ John creates a consistent chain for John. Similarly, She $\rightarrow$ Mary creates Mary's entity chain. The paragraph becomes much easier to understand when each referring expression is consistently connected to its antecedent.

c) Constraint Graph / Table

A constraint-based system can represent the resolution problem using variables and candidate domains.

Let $X_1 = \text{He}$, $X_2 = \text{She}$, $X_3 = \text{it}$, $X_4 = \text{him}$, $X_5 = \text{they}$

Candidate domains:

$D(X_1) = \{\text{John, Mary}\}$

$D(X_2) = \{\text{John, Mary}\}$

$D(X_3) = \{\text{ball, park, John, Mary}\}$

$D(X_4) = \{\text{John, Mary, dog, ball}\}$

$D(X_5) = \{\text{John, Mary, dog}\}$

Constraint Table

Variable	Candidate	Gender/Number	Recency	Semantic Compatibility	Coherence	Result
He	John	✓	✓	✓	✓	Keep
He	Mary	✗	✓	✓	✗	Eliminate
She	John	✗	✓	✓	✗	Eliminate
She	Mary	✓	✓	✓	✓	Keep
it	ball	✓	✓	✓	✓	Keep
it	park	✓	✓	✗	✗	Eliminate
him	John	✓	✓	✓	✓	Keep
him	Mary	✗	✓	✓	✗	Eliminate
him	dog	✓	✓	✗	✗	Eliminate
him	ball	✓	✓	✗	✗	Eliminate
they	John+Mary+dog	✓	✓	✓	✓	Keep

Simple Constraint Graph

He is linked to John by gender agreement and coherence. Mary is linked to She by gender agreement and coherence. For "it": ball is selected by semantic compatibility, park is rejected. For "him": John is selected by gender and semantics; Mary is rejected by gender; dog is rejected due to the self-reference problem; ball is rejected by semantic incompatibility. For "they": the group resolves to John, Mary and dog together.

d) Final Coreference Chains

The final entity chains are:

John Chain: John → He → him → they

Mary Chain: Mary → She → they

Ball Chain: ball → it

Dog Chain: dog → they

Rewritten Paragraph

To remove the ambiguity:

"John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home."

This version explicitly replaces the pronouns with their intended antecedents.

e) Priority Order of Constraints

The constraints should not necessarily have equal priority. A suitable priority order is:

1. Gender and Number Agreement
2. Semantic Compatibility
3. Coherence
4. Recency

1. Gender and Number Agreement

This is a strong grammatical constraint. For example, He → Mary can immediately be rejected because of gender mismatch.

2. Semantic Compatibility

Even if a candidate satisfies grammatical constraints, it must make semantic sense. For example, "The dog chased the ball" is possible, but "The dog chased the dog" may be grammatically possible but is not the intended interpretation in this paragraph.

3. Coherence

The interpretation must remain consistent throughout the entire paragraph. For example, He → John and him → John creates a coherent discourse.

4. Recency

Recency is useful when several candidates remain possible. However, it should not override stronger grammatical and semantic constraints. For example, dog → him could be considered because dog is recently mentioned, but semantic compatibility rejects it.

What Happens If Recency Is Relaxed?

If recency is relaxed, the system can still resolve the paragraph using gender, semantic compatibility and coherence. For example, him → John can still be selected because John is masculine, a dog can chase John, John has already been established as a discourse entity, and the resulting entity chain is coherent. However, without recency, the system may have to evaluate more candidates.

Therefore: less recency leads to more candidate links, which leads to more search, which leads to higher computational cost. Recency is therefore useful as a search-ranking constraint, but it should not override stronger linguistic constraints.

Q2. Dialog and Conversational Agents

Given Conversation

User: "I have an important exam tomorrow but I'm not able to concentrate."

The required dialog act is: Advise + Encourage

The response must maintain entity coherence, reuse "exam", "concentrate" and "you", use Cause–Effect or Elaboration, include at least two of focus, break, confident, contain only 2–3 sentences, remain logically consistent, and maintain politeness and a positive tone.

a) Discourse Planning Steps

Before generating the response, the conversational agent should perform the following steps.

Step 1 – Identify User Intent: The user is expressing a problem (unable to concentrate) in the context of an important exam tomorrow. The required response therefore needs advice and encouragement.

Step 2 – Identify Relevant Entities: Important discourse entities are exam, you, and concentrate. These should be reused to maintain entity coherence.

Step 3 – Determine Dialog Act: The dialog act is Advise + Encourage. The system should provide practical advice and motivation.

Step 4 – Select Discourse Relation: A suitable relation is Cause–Effect: because the exam is tomorrow, the student should take a short break or improve focus, so that they can study better.

Step 5 – Apply Keyword Constraints: At least two of focus, break, confident must appear.

Step 6 – Apply Length Constraint: The final response must contain exactly 2–3 sentences.

Step 7 – Apply Tone Constraints: The response should be polite, supportive, positive, and non-contradictory.

b) Three Possible Responses

Response 1: "Your exam is tomorrow, so take a short break and return with better focus instead of forcing yourself to study continuously. You can stay confident and concentrate on one topic at a time."

Response 2: "Since your exam is tomorrow, take a short break and then focus on one small topic at a time to make concentrating easier. You can feel confident because steady progress is better than stressing yourself."

Response 3: "Your exam is important, so give yourself a short break and then focus on one task at a time. You can stay confident and concentrate better by avoiding distractions and studying calmly."

c) Evaluation and Best Response

Constraint	Response 1	Response 2	Response 3
Advice	✓	✓	✓
Encourage	✓	✓	✓
Exam coherence	✓	✓	✓
Concentrate/focus	✓	✓	✓

Break	✓	✓	✓
Confident	✓	✓	✓
Cause–Effect	✓	✓	✓
2–3 sentences	✓	✓	✓
Positive tone	✓	✓	✓
Logical consistency	✓	✓	✓

Best Response: Response 2

Response 2 is the best because it provides a direct relationship between the problem and the solution: exam tomorrow leads to taking a break, which leads to focusing on one topic, which leads to improved concentration. It also naturally uses the required entities (exam, concentrating, you) and includes the required keywords (break, focus, confident). The response is concise, supportive and logically consistent.

d) Effect of Violating Two Constraints

Constraint 1 – Response Length: Suppose the system ignores the 2–3 sentence constraint and produces a long paragraph. If the user asks for simple advice and the system generates 10–15 sentences, this causes information overload and a less useful response. For a stressed student, excessive information may make the response less effective. Therefore, the length constraint improves usability.

Constraint 2 – Positive Tone: If the system violates the positive-tone constraint, it might generate: "You cannot concentrate because you are not disciplined enough." This is technically related to the problem but is discouraging and inappropriate. A conversational agent should instead provide constructive guidance such as "Take a short break and focus on one topic at a time." Therefore, politeness and positive tone are important for effective human–AI interaction.

Q3. Machine Translation, Discourse Structure and Predicate Logic

Source Sentence

"The bank by the river flooded after the storm, but it was saved by quick action."

The document explicitly identifies "bank" as ambiguous between a financial institution and a riverbank. The required tasks are word sense disambiguation, predicate logic representation, coherent

translation/paraphrase, an RST-style discourse tree, and comparison with pure statistical machine translation.

a) Word Sense Disambiguation of "Bank"

The word bank has multiple senses.

Sense 1 – Financial Institution: Example: "I deposited money in the bank." Here, bank = financial institution.

Sense 2 – Riverbank: Example: "The children sat near the bank of the river." Here, bank = land beside a river.

Context Analysis

The sentence contains "The bank by the river flooded after the storm..." The phrase "by the river" provides strong contextual evidence. A financial institution does not naturally flood as an entity located beside a river in this interpretation. A riverbank, however, can naturally flood after a storm.

Therefore: bank = RIVERBANK

Constraint-Based WSD

The candidate senses are Financial Institution and Riverbank. Applying the constraints:

Constraint	Financial Bank	Riverbank
Occurs by a river	Weak	✓
Can flood	Weak	✓
Saved by quick action	Possible but unlikely	✓
Contextual coherence	Low	High
Overall interpretation	Rejected	Selected

Therefore: Selected Sense = Riverbank

b) Predicate Logic Representation

The sentence consists of two clauses: "The bank by the river flooded after the storm" and "but it was saved by quick action." The pronoun it → bank, because the riverbank is the entity introduced in the first clause and is the entity affected by the second clause.

Define the predicates: Bank(x), River(y), Storm(s), Flood(x), Near(x,y), After(e1,e2), Saved(x,a), QuickAction(a).

We can construct Bank(b), River(r), Storm(s), QuickAction(a). The location relation is Near(b,r). The flooding event is Flood(b). The temporal relationship is After(FloodEvent, StormEvent). The saving action is Saved(b,a) with QuickAction(a).

Complete Logical Representation

$$\exists b,r,s,a [\text{RiverBank}(b) \wedge \text{River}(r) \wedge \text{Storm}(s) \wedge \text{QuickAction}(a) \wedge \text{Near}(b,r) \wedge \text{Flood}(b) \wedge \text{After}(\text{FloodEvent}, \text{StormEvent}) \wedge \text{Saved}(b,a)]$$

This representation preserves the bank, the river, the storm, flooding, the quick action, and the relationship between the events.

Coreference Representation

The pronoun "it" must be resolved: The bank $\rightarrow$ it. Therefore it = bank. The complete discourse chain is bank $\rightarrow$ it, which maintains discourse coherence.

c) Target Language / Simple English Paraphrase

A simple English paraphrase that preserves all important information is:

"The riverbank was flooded after the storm, but quick action saved it."

This is a good paraphrase because it explicitly disambiguates bank as riverbank. It preserves bank as riverbank, retains the river relationship through riverbank, and keeps flooded, storm, but, it, quick action and saved intact. No major information is lost.

Alternative Indian-Language Example – Tamil

A simple Tamil rendering could be: "புயலுக்குப் பிறகு ஆற்றங்கரை வெள்ளத்தில் மூழ்கியது, ஆனால் விரைவான நடவடிக்கையால் அது காப்பாற்றப்பட்டது." Here, ஆற்றங்கரை = riverbank. This explicitly resolves the ambiguous English word bank.

d) RST-Style Discourse Tree

The sentence contains two clauses connected by "but". The relation is Contrast.

The discourse structure is represented as a Contrast relation joining two nuclei: the first nucleus is "The bank flooded after the storm" and the second nucleus is "It was saved by quick action." The second clause

contrasts with the expected consequence of the first clause. Normally a storm leads to a flood which leads to damage, but the sentence tells us that a storm leads to a flood, which leads to quick action, which leads to the bank being saved. Thus, the Contrast relation is essential.

e) Advantages of Constraint-Based Approach over Pure Statistical Machine

Translation

A constraint-based approach is particularly useful for this sentence because it contains lexical ambiguity and a coreference relationship.

1. Explicit Word Sense Disambiguation: A statistical system may select the most common sense of "bank" based on training data. A constraint-based system explicitly evaluates bank with river, flood, and storm, and determines that riverbank satisfies the contextual constraints.

2. Better Semantic Consistency: The system can check whether the selected interpretation makes sense. Riverbank plus flood is semantically compatible, whereas financial institution plus flood is less compatible in this specific sentence.

3. Explicit Coreference Resolution: The constraint system can represent it → bank, which prevents the translation system from incorrectly assigning "it" to another entity.

4. Information Preservation: Constraint-based processing can explicitly ensure that important entities and relationships (bank, river, storm, flood, quick action, saved) are maintained, and the system can check that none of these relationships are lost during translation.

5. Discourse Relation Preservation: The word "but" indicates a Contrast relation. The system can explicitly preserve the Contrast link between Clause 1 and Clause 2, which helps prevent a translation from losing the logical relationship between the two clauses.

6. Explainability: A constraint-based system can explain why a particular interpretation was selected — for example, bank = riverbank because of the river context, flooding event, storm context, and semantic compatibility. A pure statistical system may provide an output without explicitly explaining the reasoning.

Comparison Table

Feature	Constraint-Based Approach	Pure Statistical MT
Word sense disambiguation	Explicit constraints	Probability-based
Context handling	Rule/constraint driven	Learned from data
Coreference	Explicitly represented	May be inconsistent

Predicate logic	Can represent explicitly	Usually implicit
Discourse relations	Explicit	May be lost
Explainability	High	Lower
Information preservation	Can be checked	Depends on model
Ambiguity handling	Constraint satisfaction	Probability ranking
Domain adaptation	Rules can be customized	Requires suitable data
Error analysis	Easier to trace	More difficult

Overall Constraint-Based Solution

The complete processing pipeline for the sentence is: tokenize the input sentence, identify the ambiguity in "bank", check the context for both the Financial Bank and Riverbank candidates, reject the Financial Bank sense and select Riverbank, build the predicate logic representation, resolve coreference, perform discourse analysis to identify the Contrast relation, produce the translation/paraphrase, validate against the constraints, and generate the final output.

Final Answers at a Glance

Q1 – Coreference Resolution

He → John, She → Mary, it → ball, him → John, they → John + Mary + dog.

Final rewritten paragraph: "John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home."

Q2 – Conversational Agent

Best response: "Since your exam is tomorrow, take a short break and then focus on one small topic at a time to make concentrating easier. You can feel confident because steady progress is better than stressing yourself."

It satisfies advise, encourage, exam coherence, concentrate/focus, break, confident, cause–effect, 2 sentences, positive tone, and logical consistency.

Q3 – Word Sense Disambiguation

bank → riverbank, it → bank.

Predicate representation: RiverBank(b), River(r), Storm(s), Near(b,r), Flood(b), After(FloodEvent, StormEvent), Saved(b,a), QuickAction(a).

Paraphrase: "The riverbank was flooded after the storm, but quick action saved it."

Discourse relation: Contrast, joining "Flooding after storm" and "Saved by quick action."

Conclusion

Constraint-based problem solving provides a structured way to solve difficult NLP problems by treating linguistic interpretation as a constraint satisfaction problem.

In reference resolution, grammatical agreement, recency, semantic compatibility and discourse coherence work together to determine the correct antecedent.

In conversational agents, dialog acts, discourse relations, keyword requirements, response length and politeness constraints ensure that generated responses satisfy both linguistic and conversational requirements.

In machine translation, constraints help resolve ambiguous words such as "bank," preserve coreference, maintain predicate relationships and retain discourse relations such as Contrast.

Therefore, constraint-based NLP provides an interpretable and controllable framework for producing outputs that are not only grammatically valid but also semantically correct, contextually coherent and consistent with the requirements of the application.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Develop ment	Applicati on of Concepts	Justificatio n	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

